Rangel Daroya

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EDUCATION

University of Massachusetts, Amherst, MA, USA
Ph.D. Computer Science (GPA: 4.00/4.00), September 2022 – May 2027 (expected)

- Focus: Exploring multi-task/transfer learning for multi-modal applications (e.g., text, images).
- Focus: Developing robust embeddings for large-scale environmental and scientific applications.
- *Advisor: Prof. Subhransu Maji*

University of the Philippines, Diliman, Quezon City, Philippines
M.S. Electrical Engineering (GPA: 3.95/4.00), January 2018 – July 2020
B.S. Electronics and Communications Engineering (GPA: 3.89/4.00), June 2012 – June 2017

- Summa Cum Laude & Top 2 in the entire university (3,000+ students).

WORK EXPERIENCE

University of Massachusetts, Amherst, MA, USA
Research Assistant, September 2022 – Present

- Developing novel multi-modal computer vision models that fuse satellite imagery with wildlife observation metadata to improve image embeddings (WildSAT, ICCV 2025).
- Built a global model of rivers for hydrological analysis, deployed to the NASA PO.DAAC, processing terabytes of data using HPC clusters to solve environmental problems at scale (featured in a UMass article here).

Adobe, Seattle, WA, USA
Research Intern, May 2026 – August 2026
Research Intern (Part-time), September 2026 – Present

- Developing representation learning algorithms for generative models to improve efficiency and transfer learning performance.

AngioInsight, Minnetonka, MN, USA
Engineering Intern (Part-time), June 2025 – August 2025

- Explored and implemented spatio-temporally dynamic segmentation algorithms for angiograms in medical applications.

Dolby Laboratories, San Francisco, CA, USA
Ph.D. Research Intern, May 2024 – August 2024

- Developed novel methods to explore task relationships, improving ML model performance and data efficiency.
- Resulted in a first-author paper published at CVPRW 2025 on transductive learning for segmentation models.

Thinking Machines Data Science, Taguig, Philippines [https://thinkingmachin.es/]
Team Lead & Machine Learning Researcher, February 2020 – January 2022

- Led development of a predictive model that resulted in a $2 million project with a major telecommunications company.
- Designed and deployed cloud-native data analytics applications (backend and DevOps), performing analysis on terabytes of data for an investment firm with >$300 billion in assets under management.

University of the Philippines, Quezon City, Philippines
University Researcher III, May 2018 – January 2020

- Proposed and implemented 3D reconstruction algorithms, improving over baseline methods, and resulting in a paper.
- Designed a customized algorithm for 3D semantic building map augmentation, resulting in a patent.

PUBLICATIONS (Full updated list: Google Scholar)

Journals

1. **Low Groundwater Carbon Dioxide Contributions to Stream Evasion in the United States**
   Under Review, 2026
   Elisa Friedmann, Colin J. Gleason, Rangel Daroya, Teresa Hachey, Liam Morrison, Matthew Winnick
2. **Narrow river detection with SWOT: An evaluation over the contiguous United States**
   Under Review, 2026
   Fiona B. Bennitt, Colin J. Gleason, Merritt E. Harlan, Craig B. Brinkerhoff, Kwan-Hyuck Kim, Tamlin M. Pavelsky, Lijing Wang, Elisa Friedmann, Jonathan A. Flores, Rangel Daroya
3. **Observing Earth's Hydropower Resources from Space**
   Under Review, 2026
   Jonathan Flores, Colin Gleason, Dongmei Feng, Casey Brown, Jida Wang, Konstantinos M. Andreadis, Merritt E. Harlan, Yuan Yang, Fiona B. Bennitt, Elisa Friedmann , Rangel Daroya, Hind Oubanas
4. **Operational near real time global riverine sediment flux estimates from space** [paper]
   Geophysical Research Letters (GRL), 2026.

Luisa Vieira Lucchese, Rangel Daroya, Travis T Simmons, Punwath Prum, Nikki E Tebaldi, Tamlin M Pavelsky, Subhransu Maji, John R Gardner, Colin Joseph Gleason.

5. **Improving Satellite Imagery Masking using Multi-task and Transfer Learning** [paper]
   IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS), 2025.
   Rangel Daroya, Luisa Vieira Lucchese, Travis Simmons, Punwath Prum, Tamlin Pavelsky, John Gardner, Colin Gleason, Subhransu Maji.

Refereed conferences

1. **RealBirdID: Benchmarking Bird Species Identification in the Era of MLLMs** [paper]
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
   Logan Lawrence, Oindrila Saha, Rangel Daroya, Mustafa Chasmai, Wuao Liu, Max Hamilton, Aaron Sun, Seoyun Jeong, Fabien Delattre, Subhransu Maji, Grant Van Horn.
2. **SuperRivolution: Fine-Scale Rivers from Coarse Temporal Satellite Imagery** [paper]
   IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.
   Rangel Daroya, Subhransu Maji.
3. **RiverScope: High-Resolution River Masking Dataset** [paper]
   Association for the Advancement of Artificial Intelligence (AAAI), 2026. (Oral presentation)
   AGU 2025, Remote Sensing of Rivers, Lakes, Reservoirs, and Wetlands. (Oral Presentation)
   Rangel Daroya, Taylor Rowley, Jonathan Flores, Elisa Friedmann, Fiona Bennitt, Heejin An, Travis Simmons, Marissa Jean Hughes, Camryn L Kluetmeier, Solomon Kica, J. Daniel Vélez, Sarah E. Esenther, Thomas E. Howard, Yanqi Ye, Audrey Turcotte, Colin Gleason, Subhransu Maji.
4. **WildSAT: Learning Satellite Image Representations from Wildlife Observations** [paper]
   IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
   Computer Vision for Ecology (CV4E) Workshop at ICCV, 2025. (Spotlight Talk)
   AGU 2025, Advancing Artificial Intelligence for Remote Sensing. (Oral Presentation)
   Rangel Daroya, Elijah Cole, Oisin Mac Aodha, Grant Van Horn, Subhransu Maji.
5. **Task2Box: Box Embeddings for Modeling Asymmetric Task Relationships** [paper]
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. (Highlight, 11.9% of CVPR24 papers)
   Rangel Daroya, Aaron Sun, Subhransu Maji.
6. **Alphabet sign language image classification using deep learning** [paper]
   IEEE Region 10 Conference (TENCON), 2018.
   Rangel Daroya, Daryl Perlta, Prospero Naval.
7. **NDVI image extraction of an agricultural land using an autonomous quadcopter with a filter-modified camera** [paper]
   IEEE International Conference on Control System, Computing and Engineering (ICCSCE), 2017.
   Rangel Daroya, Manuel Ramos.

Workshops

1. **T-SAM: Transductive Learning for Segment Anything Model** [paper]
   IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2025.
   Rangel Daroya, Deepak Chandran, Subhransu Maji, Andrea Fanelli.
2. **COSE: A consistency-sensitivity metric for saliency on image classification** [paper]
   IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), 2023.
   New England Computer Vision Workshop (NECV), 2022. (Oral Presentation)
   Rangel Daroya*, Aaron Sun*, Subhransu Maji. (* means equal contribution)
3. **REIN: Flexible mesh generation from point clouds** [paper]
   IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020. (Oral Presentation)
   Rangel Daroya, Rowel Atienza, Rhandley Cajote.

HONORS AND AWARDS

- AGU25 Remote Sensing Student Award for *RiverScope: High-Resolution River Masking Dataset* (2025) [page]
- PhD Portfolio Distinction from University of Massachusetts, Amherst (2024)
  - Awarded to select PhD students meeting a high standard of completion, voted by faculty.
- Paul Utgoff Memorial Graduate Scholarship in Machine Learning from University of Massachusetts, Amherst (2023)
  - Competitive scholarship awarded to a first-year graduate student in Machine Learning.
- ICCV Travel Grant for Broadening Participation (2025)
- CVPR Travel Grant for inclusion (2024)

- Oblation Scholar at University of the Philippines (2012-2017)
    - Prestigious full scholarship given to the top 50 of 60,000+ university applicants.

TEACHING EXPERIENCE

University of Massachusetts, Amherst, MA, USA
Teaching Assistant, February 2026 – Present

- Assisted with course materials and mentored students for their course projects in "COMPSCI 682 Neural Networks: A Modern Introduction"

University of the Philippines, Diliman, Quezon City, Philippines
Senior Lecturer 1, August 2021 – August 2022
Lecturer 2, January 2018 – July 2021

- Designed and taught undergraduate lectures courses on topics related to introduction to Python programming, robotics, circuits, and telecommunications.

De La Salle University, Manila, Philippines
Lecturer, March 2022 – August 2022

- Designed and taught undergraduate courses on topics related to energy conversion, AC/DC motors, and machine learning.

PATENT

- **Semantic Three-dimensional (3D) Building Augmentation,** PH12021050443 (2025). Inventors: Shakira Arguelles, Ferdinand John Briones, Izza Claire Jalandoni, Wonkyun Park, Jonric Mirando, James Carl Necio, Raimarc Dionido, Rowel Atienza, Soonyoung Lee, Rangel Daroya, Daryl Peralta.

PROFESSIONAL ACTIVITIES

- Co-organizer, 13th Workshop on Fine-Grained Visual Categorization (FGVC), CVPR 2026 [page]
- Co-organizer, Machine Learning and Friends Lunch (MLFL), UMass Amherst (2024 – present) [page]
    - MLFL is a weekly interactive forum at the university where we invite industry researchers and faculty to talk about their field of expertise
- Mentor, Early Research Scholars Program for undergraduates, UMass Amherst (2024 – present)
- Program committee and logistics committee, New England Computer Vision Workshop (NECV) 2025 [page]
- Reviewer: IJCV24, ECCV24, CVPR25, ICCV25, NeurIPS25, AAAI26, IJCV25, CVPR26, ECCV26, NeurIPS26, AAAI27
- Ph.D. Application Reviewer, UMass Amherst (2023, 2024)
- Mentor, GradMAP Philippines for STEM career mentorship (2022-2024) [page]
    - Mentorship support for students in the Philippines who want to pursue graduate studies overseas

TECHNICAL SKILLS

- Programming Languages: Python, MATLAB, C/C++
- Deep Learning: PyTorch, PyTorch-lightning, Tensorflow, scikit-learn, Vision Transformers (ViT), Self-supervised Learning
- HPC & Scalable AI: Multi-GPU Training, High Performance Computing (HPC), Docker, Shell Scripting
- Domains & Tools: Remote Sensing, Multispectral Images, NumPy, Pandas, Git, LaTeX, QGIS